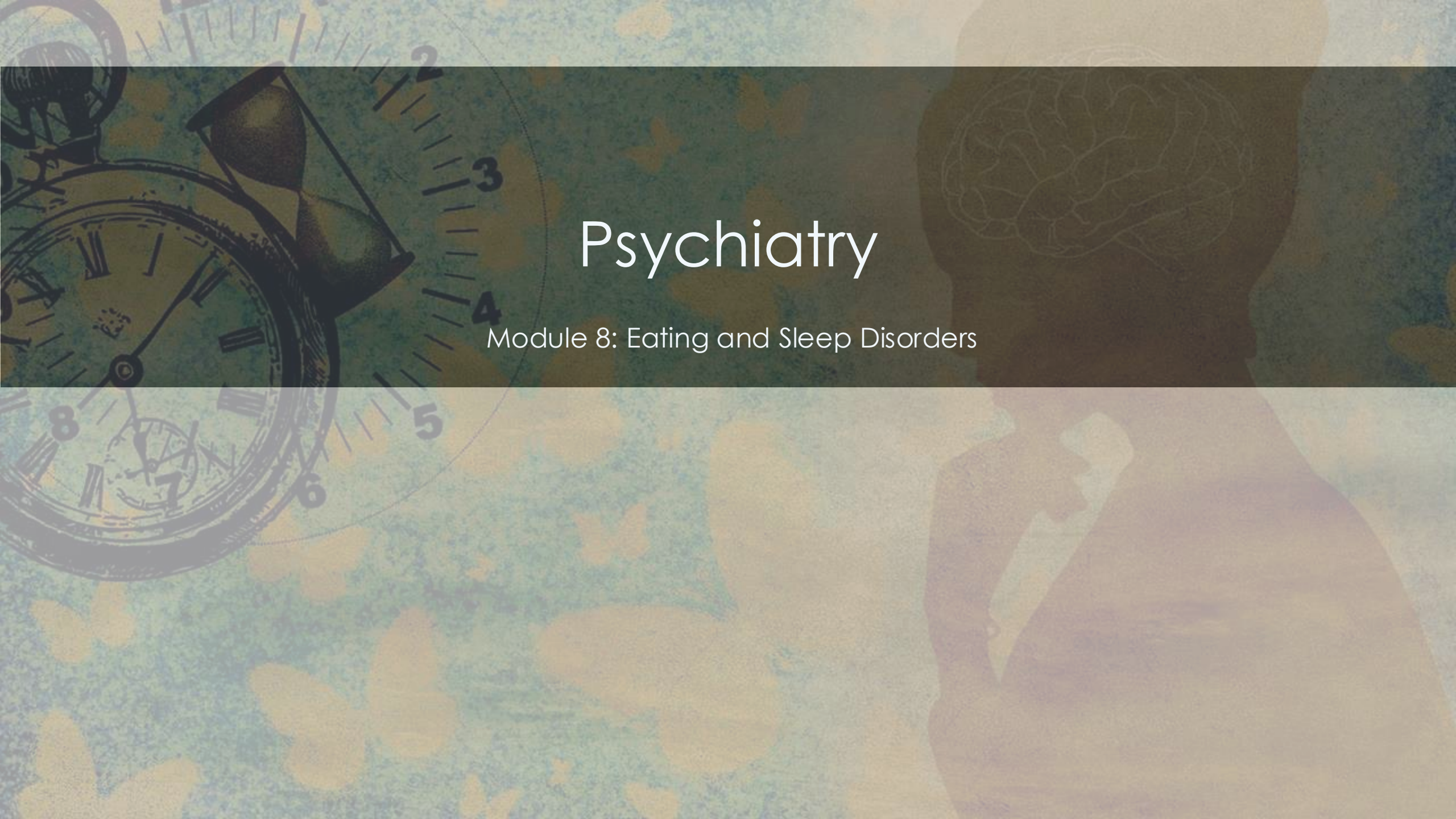


The background is a complex collage. On the left, there is a large, detailed illustration of an alarm clock with a bell and an hourglass integrated into its design. The clock face shows Roman numerals and the time is approximately 10:10. To the right of the clock, there is a faint, stylized silhouette of a human head in profile, facing right, with a brain visible inside. The entire background is covered with a pattern of soft, overlapping, organic shapes in shades of green, yellow, and brown, giving it a textured, painterly appearance.

Psychiatry

Module 8: Eating and Sleep Disorders

Module Learning Outcomes

Examine Eating and Sleep Disorders

8.1: Describe feeding and eating disorders

8.2: Distinguish between normal and abnormal sleep, and examine the characteristics, epidemiology, and etiology of sleep-wake disorders

8.3: Examine various psychological perspectives and treatment methods related to eating and sleep disorders

Feeding and Eating Disorders



Understanding Feeding and Eating Disorders

8.1: Describe feeding and eating disorders

8.1.1: Describe how hunger and eating are regulated in the human body

8.1.2: Describe anorexia nervosa, including the etiology, symptoms, and psychological health complications

8.1.3: Describe treatment options for anorexia nervosa

8.1.4: Describe the characteristics, complications, and health outcomes of bulimia nervosa

8.1.5: Explain binge eating disorder

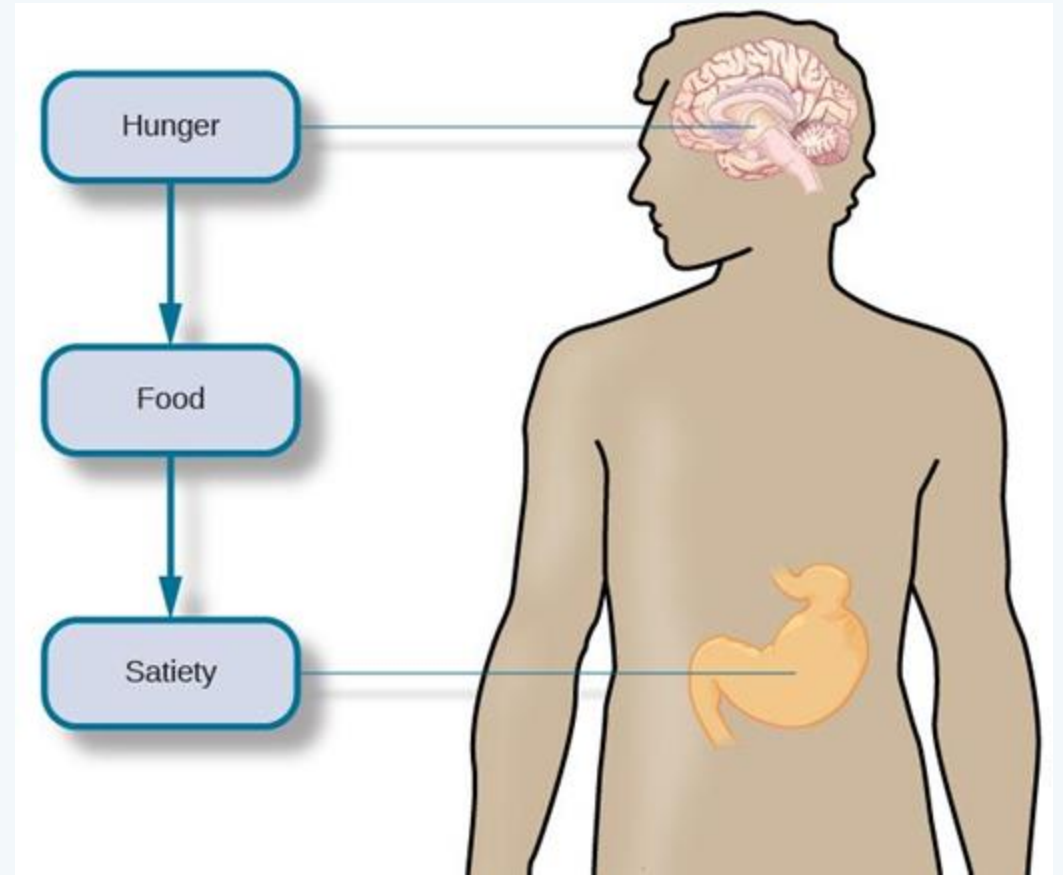
8.1.6: Describe the characteristics, complications, and health outcomes of avoidant/restrictive food intake disorder (ARFID)

8.1.7: Describe rumination disorder

8.1.8: Describe the characteristics, health outcomes, and treatment options for Pica

Hunger and Eating Regulation

- **Satiation:** fullness and satisfaction, and the eating behavior stops.
- **Leptin:** a satiety hormone
- **Metabolic rate:** the amount of energy that is expended in a given period of time.
- **Set-point theory:** each individual has an ideal body weight, or set point, which is resistant to change.
 - Body mass index (BMI) between 25 and 29.9 is considered **overweight**.
 - An adult with a BMI of 30 or higher is considered **obese**.
 - **Morbid obesity** is defined as having a BMI over 40.



Anorexia- Case Study Discussion



Class Activity: Eating Disorder PSA

Working in groups, pick an eating disorder (anorexia, bulimia, binge eating disorder, etc.) and look up its long-term effects.

- What are long-term physical problems associated with this eating disorder?

Imagine you were working for an advertising company tasked with creating a PSA to encourage people to seek treatment for eating disorders.

- What are three specific things might you mention as evidence why people should get help for the disorder?

Anorexia Nervosa

Anorexia nervosa (AN): is an eating disorder characterized by the maintenance of a bodyweight well below average through starvation and/or excessive exercise.

Symptoms include:

- Extremely restricted eating
- Extreme thinness (emaciation)
- A relentless pursuit of thinness and unwillingness to maintain a normal or healthy weight
- Intense fear of gaining weight
- Distorted body image, a self-esteem that is heavily influenced by perceptions of body weight and shape, or a denial of the seriousness of low body weight

It is estimated that approximately between 0.3 - 2 % of young women and 0.1 to 0.3 % of males will develop Anorexia.

Anorexia is a serious disease, especially given that people with anorexia aged 15 to 24 have 10 times the risk of dying compared to their similar age peers.



Treatment for Anorexia

Treatment for Anorexia is multi-dimensional and requires many different types of treatment including nutritional support, psychological counseling, and behavioral modification.

- Treatment may occur in different settings (e.g., outpatient basis, residential, or partial hospitalization) and integrates family therapy.
- Treatment may involve promoting weight gain—often an estimated amount of 2 to 3 pounds per week during inpatient care and 1 pound per week in outpatient settings.



Bulimia Nervosa

Bulimia nervosa is followed by an attempt to compensate for a large amount of consumed food.

Symptoms include:

- Chronically inflamed and sore throat
- Swollen salivary glands in the neck and jaw area
- Worn tooth enamel and increasingly sensitive and decaying teeth as a result of exposure to stomach acid
- Acid reflux disorder and other gastrointestinal problems
- Intestinal distress and irritation from laxative abuse
- Severe dehydration from purging of fluids
- Electrolyte imbalance (too low or too high levels of sodium, calcium, potassium, and other minerals) which can lead to stroke or heart attack

The lifetime prevalence rate for bulimia nervosa is estimated at around 1% for women and less than 0.5% for men



Binge Eating Disorder

Binge eating is the core symptom of **Binge Eating Disorder (BED)**; however, not everyone who binge eats has BED. An individual may occasionally binge eat without experiencing many of the negative physical, psychological, or social effects of BED.

Potential symptoms:

- Eating much faster than normal, perhaps in a short space of time
- Eating a large amount when not hungry
- Eating until feeling uncomfortably full
- Subjective loss of control over how much or what is eaten
- Binges may be planned in advance, involving the purchase of special binge foods, and the allocation of specific time for bingeing, sometimes at night
- Eating alone or secretly due to embarrassment over the amount of food consumed
- There may be a dazed mental state during the binge
- Not being able to remember what was eaten after the binge
- Feelings of guilt, shame or disgust following a food binge



Avoidant/Restrictive Food Intake Disorder (ARFID)

Avoidant/Restrictive Food Intake Disorder (ARFID) is an eating or feeding disturbance associated with an apparent lack of interest in eating or food. Disturbance in eating or feeding, is evidenced by one or more of the following:

- Substantial weight loss (or, in children, absence of expected weight gain)
- Nutritional deficiency
- Dependence on a feeding tube or dietary supplements
- Significant psychosocial interference

Stages of Treatment for Children:

- **Record stage:** children are encouraged to keep a log of their typical eating behaviors without attempting to change their habits as well as their cognitive feelings.
- **Reward stage:** involves systematic desensitization.
- **Relaxation stage:** is most important for those children that suffer severe anxiety when presented with unfavorable foods.
- **Review stage:** is important to keep track of the child's progress.



Rumination Disorder

Rumination is an eating disorder characterized by having the contents of the stomach regurgitated (drawn back up into the mouth), and either re-chewed, re-swallowed, or spit out.

- Signs and symptoms of rumination disorder include the backward flow of recently eaten food from the stomach to the mouth.
- Occurs immediately to 15-30 minutes after eating.
- May also occur following a viral illness, emotional stress, or physical injury.

The main treatment of rumination disorder is behavioral therapy.

- This may involve habit reversal strategies, relaxation, diaphragmatic breathing, and biofeedback.



Outcomes from Pica

Pica: a psychological disorder characterized by an appetite for substances that are largely non-nutritive, such as ice, soap, hair, paper, metal, soil, stones, glass, or chalk.

- Eating must persist for more than one month at an age when eating such objects is considered developmentally inappropriate, not part of a culturally sanctioned practice, and sufficiently severe to warrant clinical attention.
- Pica is most commonly seen in pregnant women, small children, iron and zinc-deficient children, malnourished children, and people with intellectual development disorders (intellectual disabilities).



Practice Question 1

Tina felt that she needed to lose some weight recently. Tina decided that she was going to start exercising as well as changing her diet. Tina altered her diet by limiting her food intake for the day at 500 calories or less. Which of these eating disorders would be consistent with the food intake?

- A. Bulimia nervosa
- B. Binge Eating Disorder
- C. Anorexia Nervosa
- D. Rumination Disorder

Sleep Disorders

Normal and Abnormal Sleep and Sleep-Wake Disorders

8.2: Distinguish between normal and abnormal sleep, and examine the characteristics, epidemiology, and etiology of sleep-wake disorders

8.2.1: Describe sleep, the sleep-wake cycle, and the stages of sleep

8.2.2: Describe symptoms and health consequences of insomnia

8.2.3: Describe symptoms and factors associated with narcolepsy

8.2.4: Describe symptoms and factors associated with hypersomnolence disorder

8.2.5: Explain NREM sleep arousal disorders (sleepwalking and night terrors)

8.2.6: Explain REM sleep behavior disorder (RBD)

8.2.7: Differentiate between parasomnias

8.2.8: Describe sleep apnea and other breathing-related sleep disorders

Sleep and the Sleep Cycle

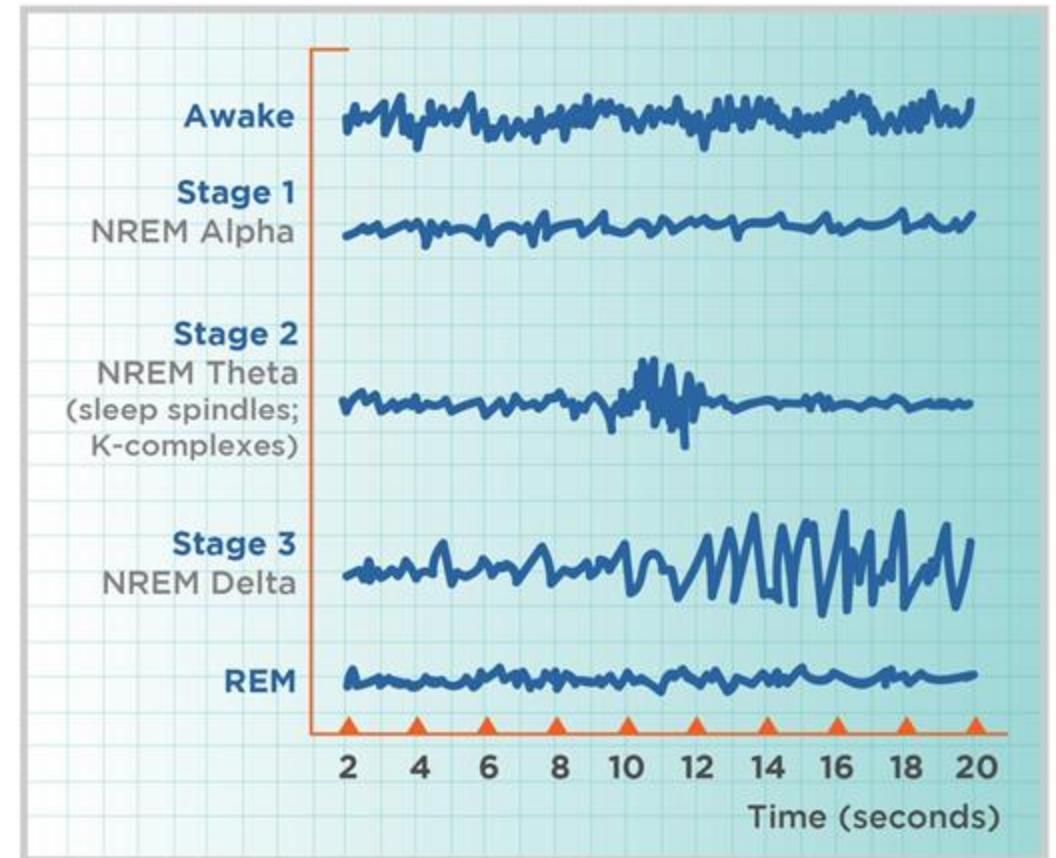
Sleep is distinguished by low levels of physical activity and reduced sensory awareness.

- Sleep-wake cycles seem to be controlled by multiple brain areas acting in conjunction with one another. Some of these areas include the thalamus, the hypothalamus, and the pons.

Stages of sleep:

- **Stage 1:** a transitional phase that occurs between wakefulness and sleep, the period during which we drift off to sleep.
- **Stage 2:** the body goes into a state of deep relaxation. Theta waves still dominate the activity of the brain, but they are interrupted by brief bursts of activity known as sleep spindles.
- **Stage 3:** often referred to as deep sleep or slow-wave sleep.

EEG RECORDINGS DURING SLEEP



Different Sleep Disorders



Insomnia

Insomnia is one of the most common sleep disorders and is characterized by a consistent difficulty in falling or staying asleep.

- Individuals often experience long delays between the times that they go to bed and actually fall asleep.
- Individuals may wake up several times during the night only to find that they have difficulty getting back to sleep.
- People suffering from insomnia tend to experience increased levels of anxiety about their inability to fall asleep.
- Can lead to a self-perpetuating cycle of insomnia because increased anxiety leads to increased arousal, and higher levels of arousal make the prospect of falling asleep even more unlikely.
- People who suffer from insomnia might limit their use of stimulant drugs (such as caffeine) or increase their amount of physical exercise during the day.



Narcolepsy

Narcolepsy: individuals experience extreme drowsiness and are unable to resist falling asleep at inopportune times.

These sleep episodes are often associated with **cataplexy**, which is a lack of muscle tone or muscle weakness, and in some cases involves complete paralysis of the voluntary muscles.

Narcolepsy with cataplexy is known as type 1 narcolepsy, while narcolepsy without cataplexy is known as type 2 narcolepsy.

These factors include:

- **Autoimmune disorders:** When cataplexy is present, the cause is most often the loss of brain cells that produce hypocretin. Although the reason for this cell loss is unknown, it appears to be linked to abnormalities in the immune system.
- **Family history:** Most cases of narcolepsy are sporadic, meaning the disorder occurs in individuals with no known family history. However, clusters in families sometimes occur—up to 10 percent of individuals diagnosed with narcolepsy with cataplexy report having a close relative with similar symptoms.
- **Brain injuries:** Rarely, narcolepsy results from traumatic injury to parts of the brain that regulate wakefulness and REM sleep or from tumors and other diseases in the same regions.

Although there is no cure for narcolepsy, some of the symptoms can be treated with medicines and lifestyle changes.



Hypersomnolence Disorder

Hypersomnolence disorder: the patient in question must have no other symptoms or signs of narcolepsy (cataplexy, sleep paralysis, hypnagogic hallucinations), and no breathing-related sleep disorder like sleep apnea.

Sleep drunkenness: Patients report waking with confusion, disorientation, slowness and repeated returns to sleep.

Treatment:

- Stimulants, such as amphetamine, methylphenidate, and modafinil, may be prescribed.
- Other drugs used to treat hypersomnia include clonidine, levodopa, bromocriptine, antidepressants, and monoamine oxidase inhibitors.



NREM Sleep Arousal Disorders

Sleepwalking: the sleeper engages in relatively complex behaviors ranging from wandering about to driving an automobile.

- During periods of sleepwalking, sleepers often have their eyes open, but they are not responsive to attempts to communicate with them.

Night terrors result in a sense of panic in the sufferer and are often accompanied by screams and attempts to escape from the immediate environment.

Nightmare disorder is a sleep disorder characterized by frequent nightmares, which often portray the individual in a situation that jeopardizes their life or personal safety.

- Studies have reported that nightmare disorders were present in 50- 70% of the cases for PTSD, in 17.5% for depression, in 18.3% for insomnia, in 16.7% for schizophrenia and in 49% for borderline personality disorder



REM Sleep Behavior Disorder (RBD)

REM sleep behavior disorder (RBD) occurs when the muscle paralysis associated with the REM sleep phase does not occur.

- These behaviors vary widely, but they can include kicking, punching, scratching, yelling, and behaving like an animal that has been frightened or attacked.
- The prevalence of RBD is approximately 0.5 to 1% in the general population and 2% in older adults.

Restless leg syndrome: experiencing uncomfortable sensations in the legs during periods of inactivity or when trying to fall asleep.

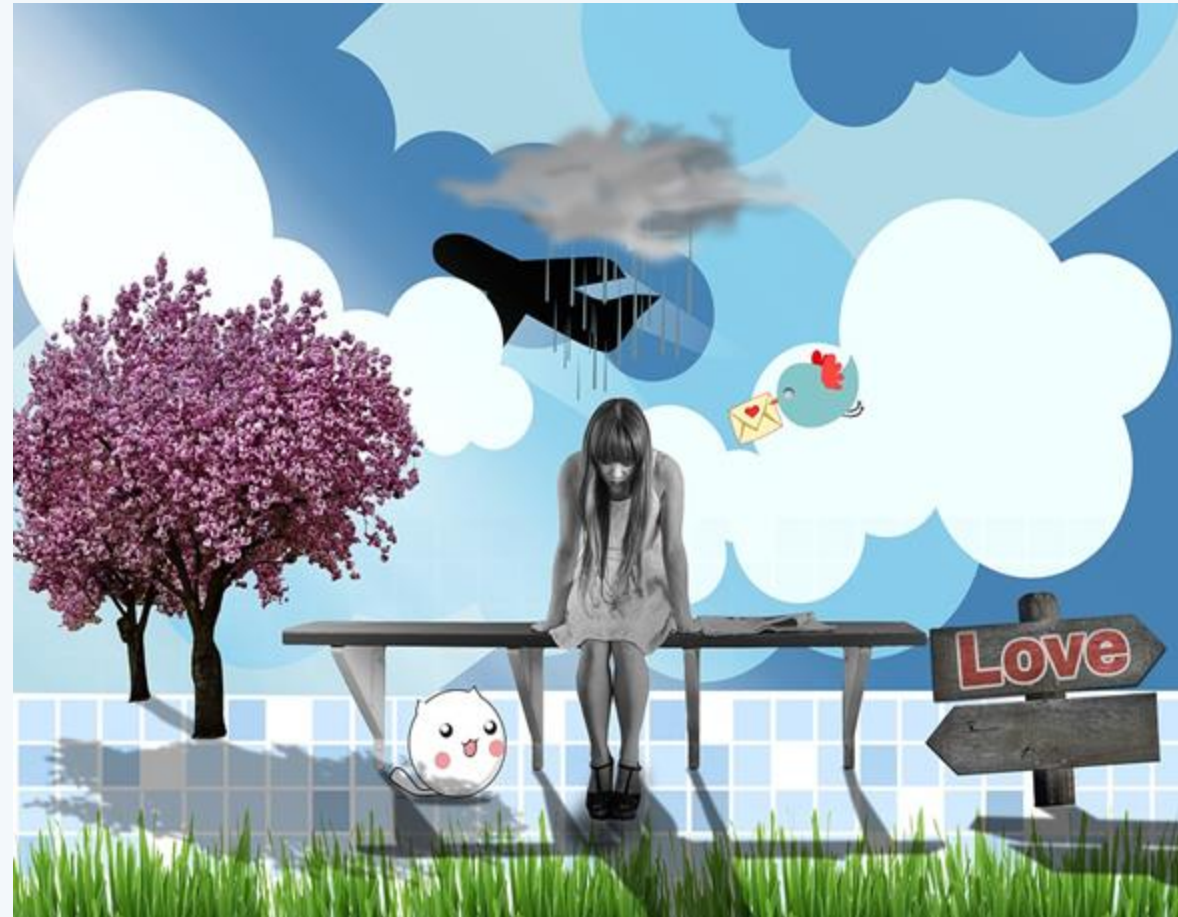
- This discomfort is relieved by deliberately moving the legs, which, not surprisingly, contributes to difficulty in falling or staying asleep.



Parasomnia Differences

A **parasomnia** is one of a group of sleep disorders in which unwanted, disruptive motor activity and/or experiences *during* sleep play a role.

- Parasomnias can occur in either REM or NREM phases of sleep.
- Parasomnias include non-rapid eye movement sleep arousal disorders of sleepwalking and sleep terrors, nightmare disorder, rapid eye movement sleep behavior disorder, and restless leg syndrome.



Sleep Apnea & Breathing Related Sleep Disorders

Sleep apnea is defined by episodes during which a sleeper's breathing stops.

- These episodes can last 10–20 seconds or longer and often are associated with brief periods of arousal.

Obstructive sleep apnea (OSA) occurs when an individual's airway becomes blocked during sleep, and air is prevented from entering the lungs.

Central sleep apnea (CSA) disrupts signals sent from the brain that regulate breathing cause periods of interrupted breathing

Sudden infant death syndrome (SIDS): an infant stops breathing during sleep and dies.

- Infants younger than 12 months appear to be at the highest risk for SIDS, and boys have a greater risk than girls.



(a)



(b)

Practice Question 2

Todd has been having trouble getting to sleep for the past few weeks. Todd has tried just about everything to get to sleep, but he still worries and is anxious that he is not able to fall asleep. Todd went to his doctor and they suggested which of the following diagnoses?

- A. Insomnia
- B. Narcolepsy
- C. RBD
- D. NREM Sleep Arousal Disorder

Perspectives on Eating and Sleep Disorders

Perspectives and Treatments of Eating and Sleep Disorders

8.3: Examine various psychological perspectives and treatment methods related to eating and sleep disorders

8.3.1: Examine eating disorders from various psychological approaches

8.3.2: Describe biological, cognitive, and behavioral approaches to treating eating disorders

8.3.3: Examine sleep disorders and sleep disorder treatment from various psychological approaches

Perspectives on Eating Disorders

Eating disorders can affect people of all ages, racial/ethnic backgrounds, body weights, and genders.

- Eating disorders frequently appear during the teen years or young adulthood but may also develop during childhood or later in life.
- Western society especially places a cultural emphasis on thinness which can influence how people view their bodies.

Other risk factors for developing eating disorders involve participating in some sports such as:

- Aesthetic sports (dance, figure skating, gymnastics) – 35%
- Weight dependent sports (judo, wrestling) – 29%
- Endurance sports (cycling, swimming, running) – 20%
- Technical sports (golf, high jumping) – 14%
- Ball game sports (volleyball, soccer) – 12%

The **psychodynamic** view on eating disorders focuses on understanding the unconscious forces and motives that influence the disorder.

Humanistic approaches take a positive approach to help the individual see themselves as more than their disorder.



Biological, Cognitive and Behavioral Approaches

- **Biological:** A genetic component related to a predisposition toward eating disorders and a genetic link has been found on chromosome 1 in multiple family members of an individual with anorexia nervosa.
- **Cognitive:** Attentional bias is the preferential attention toward certain types of information in the environment while simultaneously ignoring others.
- **Behavioral:** Eating disorders related to ways that behavior is learned and reinforced. For example, if someone who loses weight is praised or rewarded for their appearance but took extreme measures to look that way, this could contribute to the development or maintenance of an eating disorder.



Perspectives on Sleep Disorders

- **Biological:** Sleep disorders consist of medical interventions that include surgical, non-surgical, pharmacological, and nonpharmacological treatments.
- **Cognitive:** Treating sleep disorders focuses on reframing how people think about sleep and the act of sleeping to provoke change in their thoughts and sleeping habits.
- **Behavioral:** Stimulus control helps to build an association between the bedroom and sleep by limiting the type of activities allowed in the bedroom.
- **Psychodynamic:** A systematic review found that traumatic childhood experiences (such as family conflict or sexual trauma) significantly increases the risk for a number of sleep disorders in adulthood, including sleep apnea, narcolepsy, and insomnia.



Practice Question 3

Lauren recently has felt down on herself. She looked in the mirror and thought that she was too heavy, even though she only weighed 100 lbs. Lauren decided to diminish her intake of food to lose weight and she did this for about a month. One day, Lauren was walking and suddenly passed out. When she was treated, she admitted her eating disorder. Treatment would be aimed at changing her thought patterns associated with not eating. Which perspective would use this as a treatment option?

- A. Behavioral
- B. Cognitive
- C. Psychodynamic
- D. Biological

Case Study: Anorexia

Gianna was a junior in high school and felt that she was too heavy. She started to curtail her intake of food and would often eat a few bites at dinner and say that she was going to eat in the other room. Gianna also felt self-conscious around her boyfriend and was nervous eating around him as well. Gianna lost about 15 pounds in a month and felt better about herself, but was not feeling well. She had much less energy to do most things around the house and was very tired. Gianna's friends started to realize that she was not eating healthy and made sure that she received treatment. Gianna went to counseling and CBT which were effective in helping to understand why she felt bad about herself and her eating disorder.

Case Study: Binge-Eating Disorder

Elizabeth was in grad school and feeling very stressed between school and work and everything else in her life as she was away from family for most of the year. Elizabeth also felt down about herself at certain times and would eat to make herself feel better for a little bit of time, then would feel worse about it after the fact. She, at times, would eat more than 3,000 calories in about 20-30 minutes, which would only make her feel better in the short-term. After her roommate found her eating a great deal one night in the kitchen, Elizabeth realized that she needed help. She sought out a therapist that dealt specifically with BED and she recovered. Her self-esteem also increased.

Case Study: Insomnia

Jordan felt tired for the past 4 years and now that he was in college, he felt even more exhausted. Jordan would only sleep between 3-4 hours a night, waking up at 3-4am and not being able to get back to sleep. Jordan would watch videos and listen to music while he was awake, but nothing would help him fall asleep. Jordan sought out a physician who suggested sleeping pills. Jordan tried the medication, but then was unable to wake up for his classes. He then decided to visit a therapist who helped him with anxiety and stress regulation techniques such as meditation and discussed exercising more. His insomnia improved.